4.1.4.1 Manufacturing Date and Colour Test

If the cement bags stocked for long period it loses its strength. Hence, the manufacturing date of the cement bag should be checked. The colour of cement should be uniform. It is light green mixed with grey colour.

4.1.4.2 Physical Properties

When cement felt between fingers is rough, and is warm when hand is inserted in a bag of cement, it indicates adulteration.

4.1.4.3 Presence of Lumps

Cement inside the bag should not be harden due to moisture. If it is hard, such cement should be rejected.

4.1.4.4 Block Test

A cement block of size 200 mm × 25mm × 25mm is made and it is immersed in water for 7 days. It is then placed 150 mm apart between supports. The cement is good, if this block shows no sign of failure when 34kg weight is loaded on it.

4.1.4.5 Glass Plate Test

A thick paste of cement with water is made on a piece of glass plate and keep it in water for 24 hours. The paste should set hard with the glass plate when it is taken out.

4.1.5 Required Properties of Portland Cement

- Initial setting time should not be less than 30 minutes.
- Final setting time should not be more than 10 hours.
- After 3 days, compressive strength should not be less than 16 N/mm^2.
- After 7 days, compressive strength should not be less than 22 N/mm^2.
- After 3 days, tensile strength should be 2 N/mm^2.

DO YOU KNOW?

The first bridge built using artificial cement is Souillac Bridge over Dordogne River in 1824 at France.



ACTIVITY 1

Do the block test and glass plate test conducted in cement in your class room.

- After 7 days, tensile strength should be 2.5N/mm^2.
- The residue should not be more than 10% when sieved in I.S 90 micron sieve.
- Should not expand more than 19 mm in Le-Chatlier test.

4.1.6 Grades of Cement

There are three grades of cement.

1) Grade 33 as per IS 269 (1989).
2) Grade 43 as per IS 8112 (1989).
3) Grade 53 as per IS 12269 (1987).



Now a days the cement of grade 43 is easily available.

Civil_English_Unit_4.indd 83 12/21/2019 2:47:48 PM